

AMIYA MANDAL

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Greater Noida West, Delhi NCR

PROFESSIONAL SUMMARY

AI Engineer and Technical Leader with 9+ years building production AI systems across 4 domains (healthcare, insurance, telecom, enterprise). One of the rare profiles that combines management experience (Manager at PwC, Senior Manager at EXL) with deep hands-on technical delivery -- still writing production code while leading teams. Shipped 3 production Generative AI systems in the last 2 years: an autonomous LLM agent with RAG, a GPT-4 document processing pipeline, and a distributed NLP system processing 10,000 records daily. Python architect (9+ years) with LangChain, LangGraph, PyTorch, FastAPI, and cloud-native deployment on AWS/Azure. Pursuing MTech from BITS Pilani. Databricks and NVIDIA GenAI certified.

WHAT SETS ME APART

Builder + Leader	Manager title + still writes production code. Not a pure people manager, not a pure IC. Leads teams AND architects systems.
GenAI in Production	3 shipped LLM systems (not PoCs). LangGraph agents, RAG pipelines, GPT-4 integration, vector databases -- all serving real users.
4-Domain Breadth	Healthcare (Optum), Insurance (EXL), Telecom (PwC), Enterprise (Nagarro). AI techniques transfer, domain knowledge compounds.
Python + Rust	9+ years Python. Built a Rust engine (PyO3) when Python wasn't fast enough. Shows systems-level thinking beyond typical ML profiles.

CORE COMPETENCIES

Generative AI & LLM Systems	Agentic AI (LangGraph, LangChain)	RAG Pipelines & Vector Databases	Python (9+ yrs) & Rust
NLP & Deep Learning (PyTorch)	Microservices & REST APIs	Cloud (AWS, Azure)	Engineering Leadership

WORK EXPERIENCE

PwC India Feb 2026 - Present

Manager - Gurugram

- Led team to build production **AI escalation copilot**: LangGraph ReAct agent with multi-tool orchestration, hybrid RAG (PostgreSQL + ChromaDB), compliance guardrails, and FastAPI streaming backend
- Built NLP classification engine, DataAnalyst subagent (autonomous SQL + charting), and adaptive retrieval across 23 escalation types and 94 resolution playbooks
- Owned end-to-end: architecture, code, team leadership, stakeholder delivery -- concept to production

EXL Jul 2024 - Feb 2026

Senior Manager - Noida

- Led engineering team delivering **GPT-4 document processing platform**: OCR + LLM extraction, multi-stage validation, batch processing at hundreds of docs/batch with audit traceability
- Designed RAG architecture with contextual retrieval, adaptive prompting, and evaluation framework for continuous quality improvement

Nagarro Jun 2022 - May 2024

Associate Staff Engineer (Data Scientist) - Gurugram

- Built ML classification (PyTorch Transformers, 30K examples, **20% precision improvement**), PII detection (**95% accuracy**), and GPT-3 content generation integration

Optum (UnitedHealth Group)

Aug 2019 - Jun 2022

Data Scientist - Gurugram

- Built **distributed NLP pipeline** processing 10,000 healthcare calls/day: pub-sub architecture, Docker orchestration, Azure cloud, automatic failover
- Evaluated 5 Transformer models (T5, BART, Longformer), built SBERT semantic search engine, reduced transcription errors 25%. Built Rust engine (PyO3) for high-performance text processing

KiwiTech / Untrodden Labs / Secninjaz

Dec 2016 - Aug 2019

Software Engineer / Python Developer - Noida / Delhi

- Built Python web applications: Django REST APIs, PostgreSQL, MongoDB, Elasticsearch, AWS, NLP chatbot integration, social media scraping services

EDUCATION

Master of Technology (MTech) - BITS Pilani

2025 - 2027

Bachelor of Technology (Computer Science) - NIET, Greater Noida

2012 - 2016

CERTIFICATIONS

Databricks Certified Generative AI Engineer Associate

May 2025

NVIDIA Certified Associate: Generative AI LLMs

May 2025

Microsoft Career Essentials in Generative AI

Mar 2026

TECHNICAL SKILLS

Languages: Python (9+ yrs), Rust, SQL

GenAI: LangChain, LangGraph, GPT-4, OpenAI APIs, RAG, Prompt Engineering, Embeddings, Guardrails

ML/NLP: PyTorch, scikit-learn, TensorFlow, Transformers (BERT, T5, BART), SBERT, Deep Learning

Backend: FastAPI, Django, Flask, REST APIs, Microservices, Docker

Data: PostgreSQL, MongoDB, Redis, Elasticsearch, ChromaDB, Vector Databases, Pandas, NumPy

Cloud: AWS (SageMaker, S3), Azure (Batch, Blob, OpenAI), Docker, Jenkins, CI/CD